

TECHNICAL PROJECT REPORT

IMDb Sentiment Analysis Using Machine Learning and Power BI

Machine Learning Lab — BCADA 5P 126
Bachelor of Computer Applications (Data Analytics)

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Abstract

This technical report presents an end-to-end IMDb movie-review sentiment analysis workflow that combines supervised natural language processing, machine-learning classification and Power BI analytics. Review text is classified into Positive or Negative sentiment, while the reporting layer exposes predicted labels, positive-class probabilities, confidence bands and error categories. Three candidate classifiers—Logistic Regression, Linear SVM and Multinomial Naive Bayes—were compared using cross-validation. Logistic Regression recorded the strongest cross-validation accuracy and macro F1 at 90.25%. On the supplied 4,958-review test set, the final model achieved 91.09% accuracy and macro F1, ROC-AUC of 96.89%, Average Precision of 96.74% and a Brier Score of 0.0703 using a decision threshold of 0.55. The Power BI layer transforms model outputs into KPIs and review-level analytical views, including accuracy, correct/incorrect predictions, false positives, false negatives, confidence and probability measures. The project therefore demonstrates how machine-learning results can be converted into an interpretable business-intelligence reporting workflow.

Keywords: IMDb Sentiment Analysis, NLP, Text Classification, Logistic Regression, Linear SVM, Naive Bayes, Power BI, DAX

1. Introduction and Problem Definition

1.1 Project Overview

IMDb sentiment analysis is a Natural Language Processing application in which written movie reviews are transformed into structured sentiment predictions. The project connects three stages: textual input and model prediction, quantitative model evaluation, and Power BI reporting. The supplied report documents review-level actual and predicted labels, positive-class probabilities, confidence information and error categories, enabling both model assessment and dashboard analysis.

1.2 Problem Statement

The technical problem is a supervised binary classification task: given a movie-review text, determine whether its expressed sentiment is Positive or Negative. The system also produces a probability/confidence measure so that predictions can be analysed beyond a simple class label. This converts unstructured text into structured information that can be aggregated, compared and visualised.

1.3 Objectives

- Develop a binary sentiment-classification workflow for IMDb movie reviews.
- Compare multiple machine-learning classifiers using consistent cross-validation measures.
- Evaluate the final model using accuracy, macro F1, ROC-AUC, Average Precision and Brier Score.
- Generate review-level prediction and error outputs suitable for Power BI.
- Construct KPI-oriented reporting around model accuracy, sentiment distribution and prediction errors.

1.4 Scope

The project covers text-based sentiment classification, candidate-model comparison, final-model evaluation, threshold analysis, confusion-matrix interpretation and Power BI reporting. It does not claim multilingual sentiment modelling, live IMDb data collection, production-scale deployment or measured commercial impact.

2. Dataset and Data Understanding

2.1 Dataset Structure

The project uses an IMDb movie-review sentiment dataset containing review text and binary sentiment labels.

2.2 Variable Representation

Variable	Type / Encoding	Purpose
Review	Text	Input review content
Actual_Sentiment	Binary 0/1	Ground-truth target
Predicted_Sentiment	Binary 0/1	Model output
Positive_Probability	Continuous	Estimated probability of Positive
Confidence	Continuous	Confidence in predicted class
Error_Type	Categorical	Correct / False Positive / False Negative

3. Data Preparation and Exploratory Analysis

3.1 Data Cleaning

The Power BI export expects cleaned review text and corresponding sentiment labels before prediction outputs are generated. In the exported review table, review values are represented as strings and sentiment values as integers. Important steps such as duplicate-removal, outlier-removal and handling missing-values are done.

3.2 Text Preprocessing and Feature Representation

The workflow passes review text into a trained classifier and produces class predictions and positive-class probabilities. Consequently, the technical description treats preprocessing as the text-to-model representation stage without inventing vocabulary size, n-gram range or other vectoriser parameters.

3.3 Exploratory Data Analysis

The most informative supplied analytical outputs are the class distribution, candidate-model comparison and final confusion matrix. The test set is nearly balanced between Positive and Negative reviews. The model comparison shows Logistic Regression at 90.25% cross-validation accuracy, Linear SVM at 89.88%, and Multinomial Naive Bayes at 87.55%. The final confusion matrix contains 4,516 correct predictions and 442 incorrect predictions.

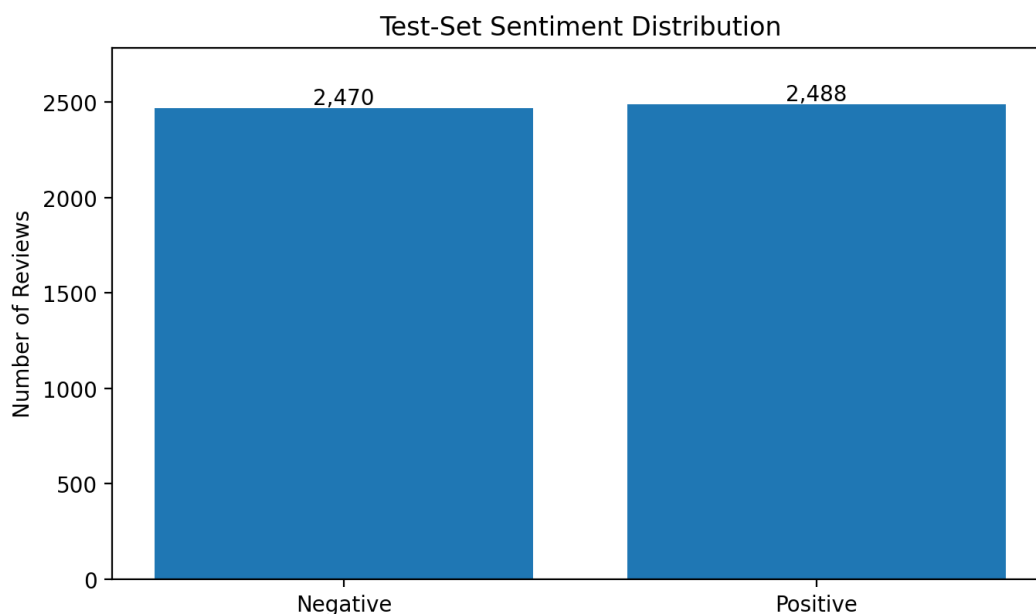


Figure 1. Actual sentiment distribution in the supplied 4,958-review test set.

3.4 Key EDA Findings

- The evaluated test set is almost evenly divided between Negative and Positive reviews.
- Logistic Regression has the highest recorded cross-validation accuracy and macro F1 among the three tested models.
- False positives and false negatives are of similar magnitude, indicating errors occur in both directions.
- Confidence and error-category fields enable review-level investigation in Power BI.

4. Machine Learning Methodology

4.1 Learning Problem

The task is supervised binary classification applied to textual data. The target classes are Positive and Negative.

4.2 Validation Strategy

Candidate models were evaluated using cross-validation, followed by final evaluation on a separate test set of 4,958 reviews. The Split ratio of the data is 80:20.

4. Machine Learning Methodology — Continued

4.3 Algorithms Evaluated

Model	CV Accuracy	CV Macro F1
Logistic Regression	90.25%	90.25%
Linear SVM	89.88%	89.88%
Multinomial Naive Bayes	87.55%	87.55%

The three models provide a comparison of linear discriminative and probabilistic approaches for text classification..

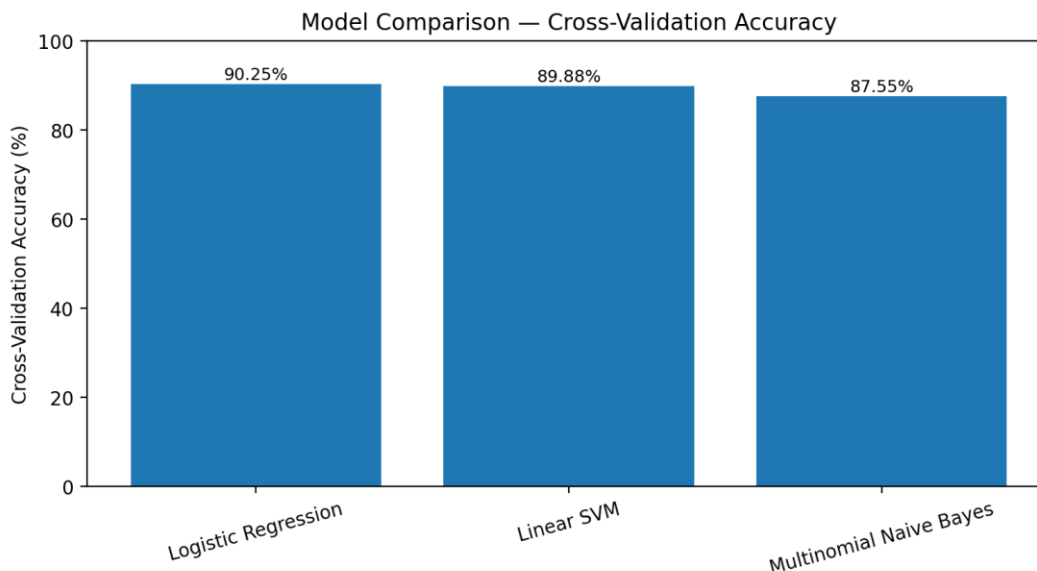


Figure 2. Cross-validation accuracy comparison of the tested classifiers.

4.4 Training and Prediction Workflow

The analytical pipeline generates candidate-model predictions for comparison. Logistic Regression is then used for final test-set prediction and positive-class probability estimation. The Power BI export consumes `X_test`, `y_test`, `test_pred`, `test_prob` and `final_metrics` to create review-level predictions, model-comparison data, final performance metrics, confusion-matrix records, threshold-analysis data, error-analysis data and dataset summaries. The documented implementation uses Python, NumPy, pandas and scikit-learn, followed by Power BI and DAX for reporting.

4.5 Threshold Analysis

The documented tuning activity is decision-threshold selection rather than a fully specified hyperparameter search. The final decision threshold is 0.55. Threshold-analysis data compare candidate thresholds with the selected `best_threshold` value, allowing the effect of the decision boundary to be inspected.

4.6 Evaluation Concepts

- Accuracy = $(TP + TN) / (TP + TN + FP + FN)$, measuring the proportion of correct classifications.
- Precision = $TP / (TP + FP)$, measuring how many predicted Positive cases are actually Positive.
- Recall = $TP / (TP + FN)$, measuring how many actual Positive cases are detected.
- F1-score balances precision and recall; macro F1 averages class-wise F1 values equally.
- ROC-AUC measures ranking/separation performance across thresholds; Average Precision summarises the precision-recall relationship.
- Brier Score measures the mean squared error of probabilistic predictions, with lower values indicating better probability calibration.

5. Model Evaluation and Results

5.1 Final Model Performance

Metric	Recorded Result
Test Accuracy	91.09%
Test Macro F1	91.09%
ROC-AUC	96.89%
Average Precision	96.74%
Brier Score	0.0703
Decision Threshold	0.55

5.2 Confusion Matrix and Error Analysis

The final Logistic Regression confusion matrix contains 2,259 True Negatives, 211 False Positives, 231 False Negatives and 2,257 True Positives. Therefore, 4,516 of 4,958 test predictions were correct and 442 were incorrect. The two error types are close in count, showing that the model does not exhibit a large imbalance between positive-direction and negative-direction mistakes.

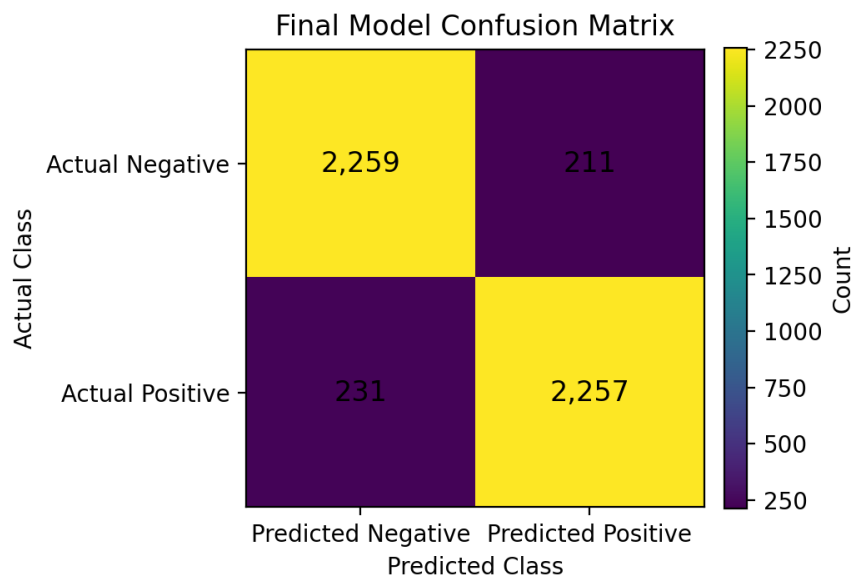


Figure 3. Confusion matrix for the final Logistic Regression model.

5.3 Derived Class Metrics

From the supplied confusion-matrix counts, Positive precision is approximately 91.45% and Positive recall approximately 90.72%. Negative precision is approximately 90.72% and Negative recall approximately 91.46%. The resulting macro F1 is approximately 91.09%, consistent with the reported final metric.

5.4 Interpretation

The final model correctly identifies most reviews in both sentiment classes and shows strong ranking performance in the evaluated test output. ROC-AUC of 0.9689 and Average Precision of 0.9674 indicate strong separation of the classes under the evaluated conditions. The Brier Score of 0.0703 provides a probability-quality measure. However, the 442 incorrect predictions demonstrate that ambiguous or context-dependent reviews remain challenging.

5.5 Model Selection

Logistic Regression was selected as the final model because it recorded the strongest cross-validation accuracy and macro F1 among the three tested classifiers, at 90.25% for both measures. Its final test evaluation is the most complete among the supplied model artifacts, including accuracy, macro F1, ROC-AUC, Average Precision, Brier Score and threshold information.

6. Power BI Dashboard and Implementation

6.1 Analytical Dashboard Layer

The Power BI layer is designed as an analytical reporting surface over review-level predictions and model-performance exports. Each review record can expose actual sentiment, predicted sentiment, positive probability, confidence, confidence band and error type. This structure supports filtering and segmentation rather than presenting only a single model-accuracy number.

6.2 KPI Measures

- Total Reviews — number of evaluated review records.
- Accuracy — calculated as Correct Predictions divided by Total Reviews.
- Positive Reviews and Negative Reviews — class-level distribution measures.
- False Positives and False Negatives — error-direction measures.
- Average Confidence and Average Positive Probability — probability-oriented model outputs.

6.3 Prediction-to-Dashboard Workflow

Review Text	Text Representation	Classifier	Probability	Threshold 0.55	Power BI KPIs
The implemented flow is: review text → text preparation / feature representation → trained classifier → positive-class probability → decision threshold 0.55 → Positive/Negative prediction and confidence → Power BI export → KPI, model and error visualisation. The export is intentionally downstream of the corrected ML pipeline so that dashboard values are based on generated model outputs.					

7. Challenges, Limitations and Future Improvements

7.1 Challenges

The project required coordination across data preparation, model comparison, evaluation, prediction export and dashboard construction. A further technical challenge is the nature of sentiment text itself: natural-language expressions may contain ambiguity, sarcasm, contextual meaning and domain-specific wording. The workflow therefore needs both model evaluation and error analysis to understand where predictions fail.

7.2 Current Limitations

- A full hyperparameter-search configuration is not supplied; documented tuning focuses on the decision threshold.
- The original dataset source URL and access date are not included.
- Text classification can remain sensitive to sarcasm, context-dependent language and ambiguous expressions.

7.3 Future Improvements

- Perform systematic hyperparameter tuning and document reproducible search settings.
- Evaluate richer text representations and compare them against the current pipeline.
- Extend calibration and threshold-sensitivity analysis.
- Create more detailed error categories for difficult review types.
- Use larger or more diverse evaluation data and monitor prediction drift.
- Add Power BI drill-through to individual reviews and time-based monitoring when timestamps are available.
- For production deployment, introduce an API/application layer, model versioning, monitoring and controlled retraining.

8. Team Contributions and Learning Outcomes

The six-member team was organised into complementary roles. Jerome handled data engineering and structured data movement; Dharshan supported analysis and interpretation; Prathik worked on model comparison and evaluation; Diganth managed the machine-learning prediction workflow; Rakshith supported analytical structuring; and Chandan prepared the Power BI export, DAX measures and dashboard-oriented structures. The project provided hands-on experience in data handling, NLP-oriented machine learning, model evaluation, confusion-matrix analysis, threshold interpretation, Power BI preparation and DAX KPI construction.

9. Conclusion

The project demonstrates an end-to-end IMDb sentiment classification and analytics workflow. Logistic Regression produced the strongest cross-validation result among the tested models at 90.25% accuracy and macro F1. Its final test evaluation achieved 91.09% accuracy and macro F1, ROC-AUC of 96.89%, Average Precision of 96.74%, Brier Score of 0.0703 and a decision threshold of 0.55. Across 4,958 test reviews, 4,516 predictions were correct and 442 were incorrect. The Power BI layer extends these machine-learning results into review-level KPIs, confidence analysis and error reporting. The central technical outcome is the integration of machine-learning evaluation with structured business-intelligence reporting while keeping conclusions tied to the available project evidence.

10. References

- IMDb movie-review sentiment dataset used by the project — source URL and access date not supplied in the project artifacts.
- Project Python / scikit-learn model pipeline and Power BI export artifacts supplied with the report materials.
- IMDb Sentiment Analysis — Power BI DAX measures supplied with the project artifacts.
- IMDb Sentiment — Dark Analytics Power BI theme supplied with the project artifacts.

Source basis: condensed and technically reorganised from the supplied 15-page project report. The source report identifies the project title, team, dataset outputs, model metrics, confusion matrix, Power BI structures and limitations. [filecite turn0file0L61-L78](#)